

SupplyGuard AI:

AI-Powered Supply Chain Resilience & Fleet Optimization

Team TechSquad || AI Track || IBM Bob AI Innovation Hackathon 2026

THE PROBLEM

Critical gaps disrupting global supply chains



FREQUENT DISRUPTIONS

Weather, port congestion, and geopolitical events delay shipments



UNDERUTILIZED FLEETS

Idle trucks on some routes while others face delays



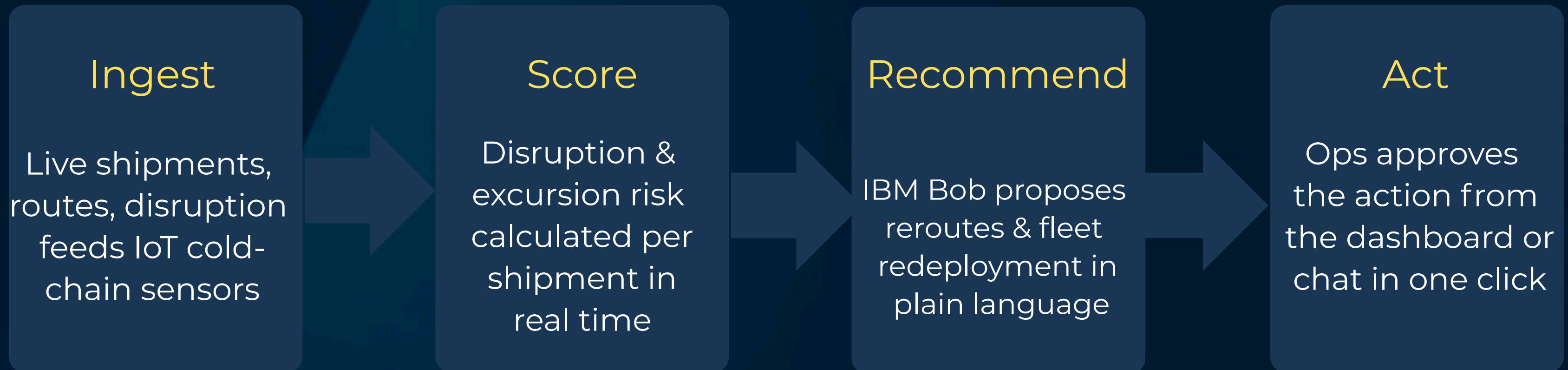
COLD-CHAIN RISK

Pharma and food shipments spoil due to undetected temperature breaches

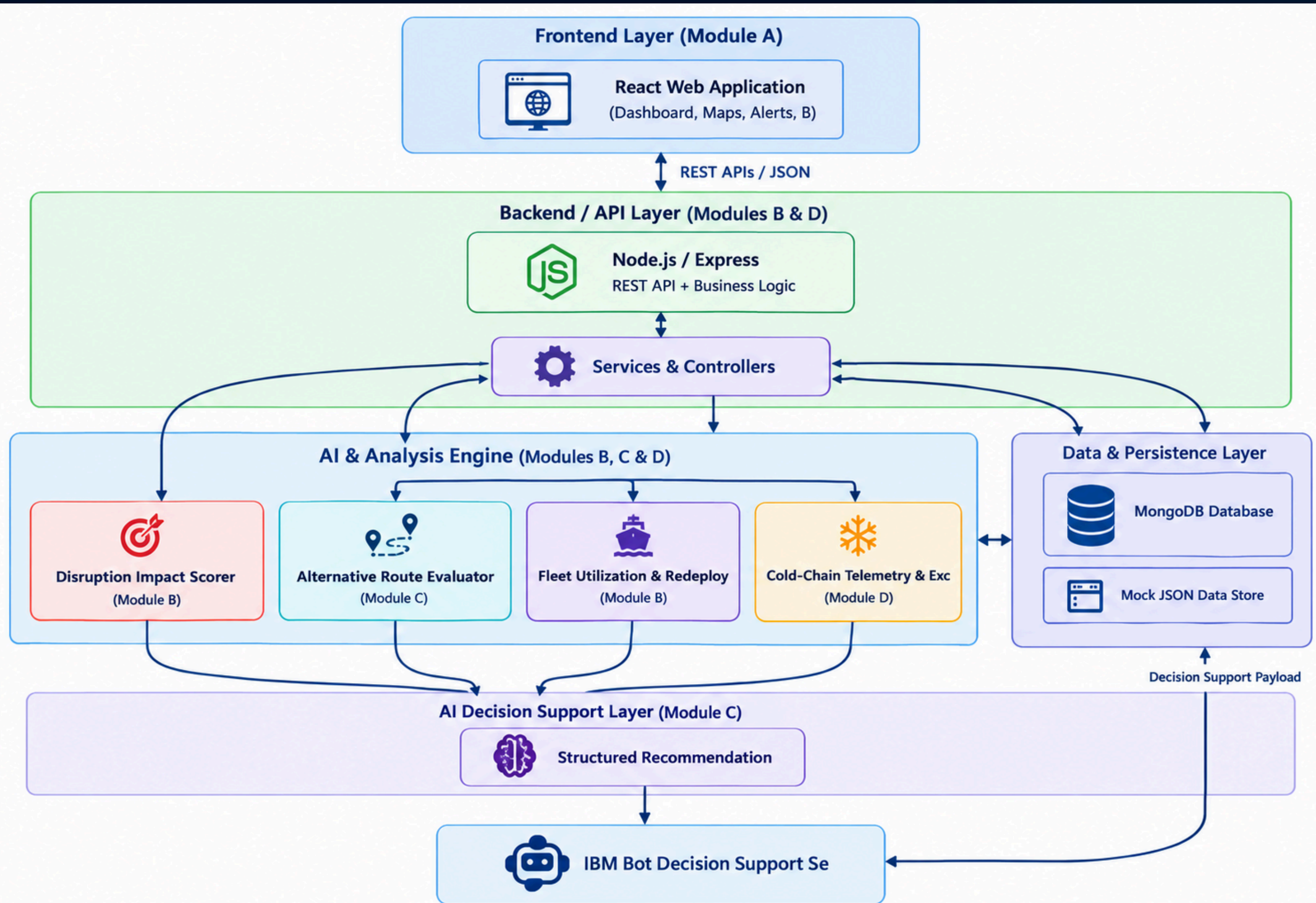
OUR SOLUTION

SupplyGuard AI Real-Time Decision Support

SupplyGuard AI is a real-time decision-support platform that watches live disruption feeds and shipment data, scores risk the moment it appears, and recommends the reroute or fleet redeployment needed to stay ahead of the problem — all through a conversational assistant powered by IBM Bob.



ARCHITECTURE



DEMO & KEY FEATURES

See SupplyGuard AI in action

Disruption Scorer

Flags impacted shipments in real time

01

Fleet Redeployment

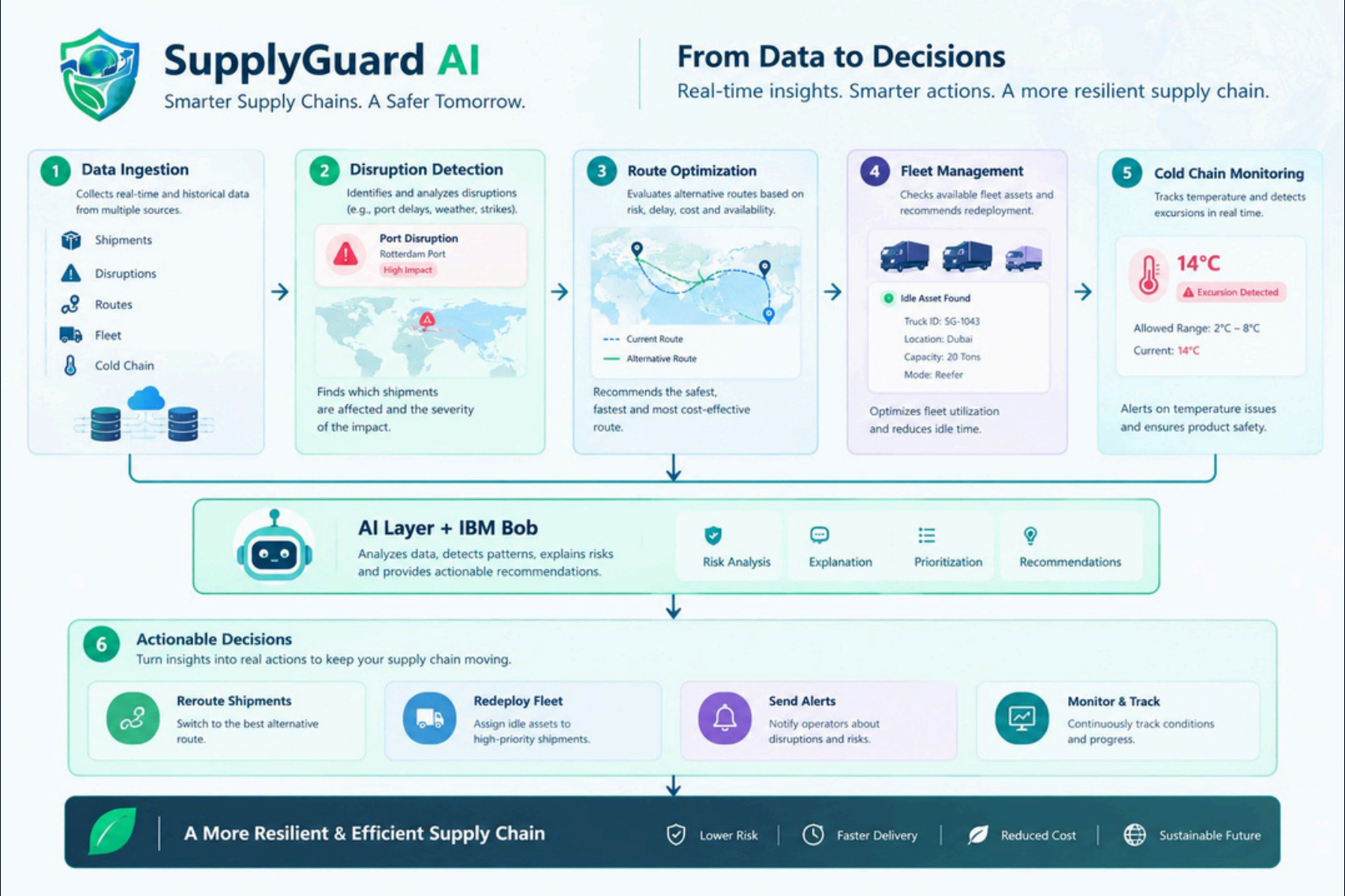
Optimizes asset reassignment

02

Cold-Chain ExcursionDetector

Alerts on temperature breaches

03



01

Disruption hits

Weather, ports, or events occur

→

02

Scorer flags shipments

SupplyGuard AI identifies high-risk shipments

→

03

IBM Bob recommends actions

AI suggests reculates, modes, and asset reassignments

→

04

Operations team acts

Approves and executes recommended actions

Powered by IBM Bob AI

Conversational AI for smarter
supply chain decisions

Answers operational questions

Explains risk factors

Drafts recommendations

Summarizes briefings



IBM Bob
AI for business



IBM Bob

● Online

Ops Manager

Why are shipments to the Northeast
delayed this week?

10:28 AM

IBM Bob

Severe weather at key ports and a backlog of
containers are causing delays on routes to
the Northeast.

10:28 AM

Ops Manager

What are the top risk factors we should watch?

10:29 AM

IBM Bob

Adverse weather, port congestion, and equipment
shortages are the primary risk factors.

10:29 AM

Ops Manager

Draft actions we can take to mitigate these risks.

10:30 AM

IBM Bob

Reroute shipments, adjust inventory buffers,
and monitor port conditions in real time.

10:30 AM

IMPACT & FUTURE VISION

Delivering outcomes today. Building the future of resilient supply chains.

FASTER **RESPONSE**

Quicker disruption handling
and issue resolution

PROTECTED **CARGO**

Reduced spoilage
and shipment losses

FULLER **FLEETS**

Improved utilization and
operating efficiency

BEYOND THE HACKATHON

Multi-tenant SaaS platform

Predictive risk modeling enhancements

Deeper integration with watsonx.ai

Fleet marketplace for asset sharing



Team TechSquad

FrontendDevelope –Priyanshi Patel

Building intuitive, responsive experiences

Shipment & Disruption Data Specialist–Drashti Dedaniya

Turning data into insights and early warnings

AI / IBM Bob Integration Engineer–Palak Bhut

Connecting AI capabilities to real-world workflows

Fleet & Cold-Chain Logic Expert–Niyati Barochia

Designing smarter routes and protecting every degree



THANK YOU

Together, we're building resilient,
intelligent supply chains.